

had of being tarred and feathered as a heretic on the University of Chicago quadrangle.

The idea behind efficient markets actually got its start in the 1800s when Charles Dow (founder of Dow Jones & Co. and the *Wall Street Journal*) commented on the market as an efficient processor of information: “In fact, the market reduces to a bloodless verdict all knowledge bearing on finance, both domestic and foreign. The price movements, therefore, represent everything everybody knows, hopes, believes, and anticipates.” In his 1889 book called *The Stock Markets of London, Paris, and New York*, George Gibson wrote, “Shares become publicly known in an open market. The value which they acquire may be regarded as the judgment of the best intelligence concerning them.” Followers of EMH later claimed as their own the concept that prices reflect all available public information.

A more substantive rationale for EMH appeared in the PhD thesis of Louis Bachelier in 1900. Bachelier compared the behavior of stock market buyers and sellers to the random movement of particles suspended in fluid. Bachelier concluded that stock price movements are random, and it is impossible to make predictions about them. Prior to this, in 1863, Jules Regnault used a randomness model to say that the deviation of stock prices is directly proportional to the square root of time. Bachelier, however, was the first to accurately model stochastic processes. These were called Brownian motion, after the Scottish botanist Robert Brown, who in 1826 first noted the random movements of pollen grains suspended in water. Einstein got the credit for explaining Brownian motion mathematically in 1905, but Bachelier had already done so in his PhD dissertation written five years earlier. Bachelier was also way ahead of his time with respect to his pioneering work on probability theory.³

Bachelier turned his PhD thesis into a book published in 1900 called *The Theory of Speculation*. It did not attract much attention until the statistician Leonard “Jimmy” Savage rediscovered it while doing research on the history of probability. Savage was so impressed with Bachelier’s pioneering work with respect to speculative markets that he sent postcards about it to a dozen or so economists he knew in the mid-1950s.

Paul Samuelson had been working on similar ideas himself. He was delighted to hear from Savage and find out about Bachelier. It allowed Samuelson to put all the pieces together into a unified equilibrium